

Recap Lecture-2



Kleene Star Closure, Plus operation, recursive definition of languages, INTEGER, EVEN, factorial, PALINDROME, $\{a^n b^n\}$, languages of strings (i) ending in a, (ii) beginning and ending in same letters, (iii) containing aa or bb (iv) containing exactly aa,

Task

Q)

- 1) Let $S = \{ab, bb\}$ and $T = \{ab, bb, bbbb\}$ Show that $S^* = T^*$ [Hint $S^* \subseteq T^*$ and $T^* \subseteq S^*$]
- 2) Let $S = \{ab, bb\}$ and $T = \{ab, bb, bbb\}$ Show that $S^* \neq T^*$ But $S^* \subset T^*$

Solution: Since $S \subset T$, so every string belonging to S^* , also belongs to T^* but bbb is a string belongs to T^* but does not belong to S^* .



3) Let $S = \{a, bb, bab, abaab\}$ be a set of strings. Are $abbabaabab$ and $baabbbabbaabb$ in S^* ? Does any word in S^* have odd number of b's?

Solution: since $abbabaabab$ can be grouped as $(a)(bb)(abaab)ab$, which shows that the last member of the group does not belong to S , so $abbabaabab$ is not in S^* , while $baabbbabbaabb$ can not be grouped as members of S , hence $baabbbabbaabb$ is not in S^* . Since each string in S has even number of b's so there is no possibility of any string with odd number of b's to be in S^* .

Task

Q1) Is there any case when S^+ contains Λ ? If yes then justify your answer.

Solution: consider $S = \{\Lambda, a\}$ then

$$S^+ = \{\Lambda, a, aa, aaa, \dots\}$$

Here Λ is in S^+ as member of S . Thus Λ will be in S^+ , in this case.



Q2) Prove that for any set of strings S

i. $(S^+)^* = (S^*)^*$

Solution: In general Λ is not in S^+ , while Λ does belong to S^* . Obviously Λ will now be in $(S^+)^*$, while $(S^*)^*$ and S^* generate the same set of strings. Hence $(S^+)^* = (S^*)^*$.

Q2) continued...

ii) $(S^+)^+ = S^+$

Solution: since S^+ generates all possible strings that can be obtained by concatenating the strings of S , so $(S^+)^+$ generates all possible strings that can be obtained by concatenating the strings of S^+ , will not generate any new string.

Hence $(S^+)^+ = S^+$

Q2) continued...

iii) Is $(S^*)^+ = (S^+)^*$

Solution: since Λ belongs to S^* , so Λ will belong to $(S^*)^+$ as member of S^* . Moreover Λ may not belong to S^+ , in general, while Λ will automatically belong to $(S^+)^*$.

Hence $(S^*)^+ = (S^+)^*$

Regular Expression

As discussed earlier that a^* generates $\Lambda, a, aa, aaa, \dots$

and a^+ generates $a, aa, aaa, aaaa, \dots$, so the language $L_1 = \{\Lambda, a, aa, aaa, \dots\}$ and $L_2 = \{a, aa, aaa, aaaa, \dots\}$ can simply be expressed by a^* and a^+ , respectively.

a^* and a^+ are called the regular expressions (RE) for L_1 and L_2 respectively.

Note: a^+ , aa^* and a^*a generate L_2 .

Recursive definition of Regular Expression(RE)

Step 1: Every letter of Σ including Λ is a regular expression.

Step 2: If r_1 and r_2 are regular expressions then

1. (r_1)

2. $r_1 r_2$

3. $r_1 + r_2$ and

4. r_1^*

are also regular expressions.

Defining Languages (continued)...

Method 3 (Regular Expressions)

Consider the language $L = \{\Lambda, x, xx, xxx, \dots\}$ of strings, defined over $\Sigma = \{x\}$.

We can write this language as the Kleene star closure of alphabet Σ or $L = \Sigma^* = \{x\}^*$

this language can also be expressed by the regular expression x^* .

Similarly the language $L = \{x, xx, xxx, \dots\}$, defined over $\Sigma = \{x\}$, can be expressed by the regular expression x^+ .



Now consider another language L , consisting of all possible strings, defined over $\Sigma = \{a, b\}$. This language can also be expressed by the regular expression

$$(a + b)^*.$$

Now consider another language L , of strings having exactly double a , defined over $\Sigma = \{a, b\}$, then it's regular expression may be

$$b^* a a b^*$$



Now consider another language L, of even length, defined over $\Sigma = \{a, b\}$, then it's regular expression may be

$$((a+b)(a+b))^*$$

Now consider another language L, of odd length, defined over $\Sigma = \{a, b\}$, then it's regular expression may be

$$(a+b)((a+b)(a+b))^* \text{ or } ((a+b)(a+b))^*(a+b)$$

Remark



It may be noted that a language may be expressed by more than one regular expressions, while given a regular expression there exist a unique language generated by that regular expression.



Example:

Consider the language, defined over $\Sigma = \{a, b\}$ of words having at least one a , may be expressed by a regular expression $(a+b)^*a(a+b)^*$.

Consider the language, defined over $\Sigma = \{a, b\}$ of words having at least one a and one b , may be expressed by a regular expression

$$(a+b)^*a(a+b)^*b(a+b)^* + (a+b)^*b(a+b)^*a(a+b)^*$$



Consider the language, defined over $\Sigma = \{a, b\}$, of words starting with double a and ending in double b then its regular expression may be $aa(a+b)^*bb$

Consider the language, defined over $\Sigma = \{a, b\}$ of words starting with a and ending in b OR starting with b and ending in a, then its regular expression may be $a(a+b)^*b + b(a+b)^*a$

TASK

Consider the language, defined over $\Sigma=\{a, b\}$ of **words beginning with a**, then its regular expression may be $a(a+b)^*$

Consider the language, defined over $\Sigma=\{a, b\}$ of **words beginning and ending in same letter**, then its regular expression may be $(a+b)+a(a+b)^*a+b(a+b)^*b$

TASK

Consider the language, defined over $\Sigma=\{a, b\}$ of **words ending in b**, then its regular expression may be $(a+b)^*b$.

Consider the language, defined over $\Sigma=\{a, b\}$ of **words not ending in a**, then its regular expression may be $(a+b)^*b + \Lambda$. It is to be noted that this language may also be expressed by $((a+b)^*b)^*$.

SummingUP Lecture 3

RE, Recursive definition of RE, defining languages by RE, $\{x\}^*$, $\{x\}^+$, $\{a+b\}^*$, Language of strings having **exactly one aa**, Language of strings of **even length**, Language of strings of **odd length**, RE defines unique language (as Remark), Language of strings having **at least one a**, Language of strings havgin **at least one a and one b**, Language of strings **starting with aa and ending in bb**, Language of strings **starting with and ending in different letters**.